第3章作业

P78：第6题

$a_1 = \frac{f(0)-f(0.5\pi)}{0-0.5\pi} = \frac{2}{\pi} \approx 0.636620$
$f'(x_2) = \cos x_2 = a_1 \Rightarrow x_2 = \arg\cos\frac{2}{\pi} \approx 0.880689$
$a_0 = \frac{f(0)+f(x_2)}{2} - \frac{x_2+0}{2} \cdot \frac{f(0)-f(0.5\pi)}{0-0.5\pi} \approx 0.105257$

故要求的最佳一次逼近多项式为：

$$P_1(x) = 0.105257 + 0.636620x$$

其误差$\max\limits_{0\le x\le 0.5\pi} |\sin x - P_1(x)| = 0.105257$

P78：第17题

$f(x) = x^2, x \in [0,1]$，设权函数$\rho(x) \equiv 1$
先看φ_1
其法方程为

$$\begin{cases} a_0(1,1) + a_1(1,x) = (x^2,1) \\ a_0(x,1) + a_1(x,x) = (x^2,x) \end{cases}$$

解得$a_0 = -\frac{1}{6}, a_1 = 1$
故最佳平方逼近$S^*(x) = a_0 \cdot 1 + a_1 \cdot x = -\frac{1}{6} + x$

再看φ_2
其法方程为

$$\begin{cases} a_0(x^{100},x^{100}) + a_1(x^{100},x^{101}) = (x^2,x^{100}) \\ a_0(x^{101},x^{100}) + a_1(x^{101},x^{101}) = (x^2,x^{101}) \end{cases}$$

解得$a_0 = 375.242532, a_1 = -375.148245$
故最佳平方逼近$S^*(x) = a_0 \cdot x^{100} + a_1 \cdot x^{101} = 375.242532x^{100} - 375.148245x^{101}$

ϕ_1上得到的最佳平方逼近更好

P78：第20题（Chebyshev多项式展开不做）

$f(x)=\sin\frac{1}{2}x$，设权函数$\rho(x)\equiv 1$

$$
\begin{aligned}
(f,P_0)&=\int_{-1}^{1}\sin\frac{1}{2}xdx=0\\
(f,P_1)&=\int_{-1}^{1}x\sin\frac{1}{2}xdx=0.325074\\
(f,P_2)&=\int_{-1}^{1}(\frac{3}{2}x^2-\frac{1}{2})\sin\frac{1}{2}xdx=0\\
(f,P_3)&=\int_{-1}^{1}(\frac{5}{2}x^3-\frac{3}{2}x)\sin\frac{1}{2}xdx=-0.002348
\end{aligned}
$$

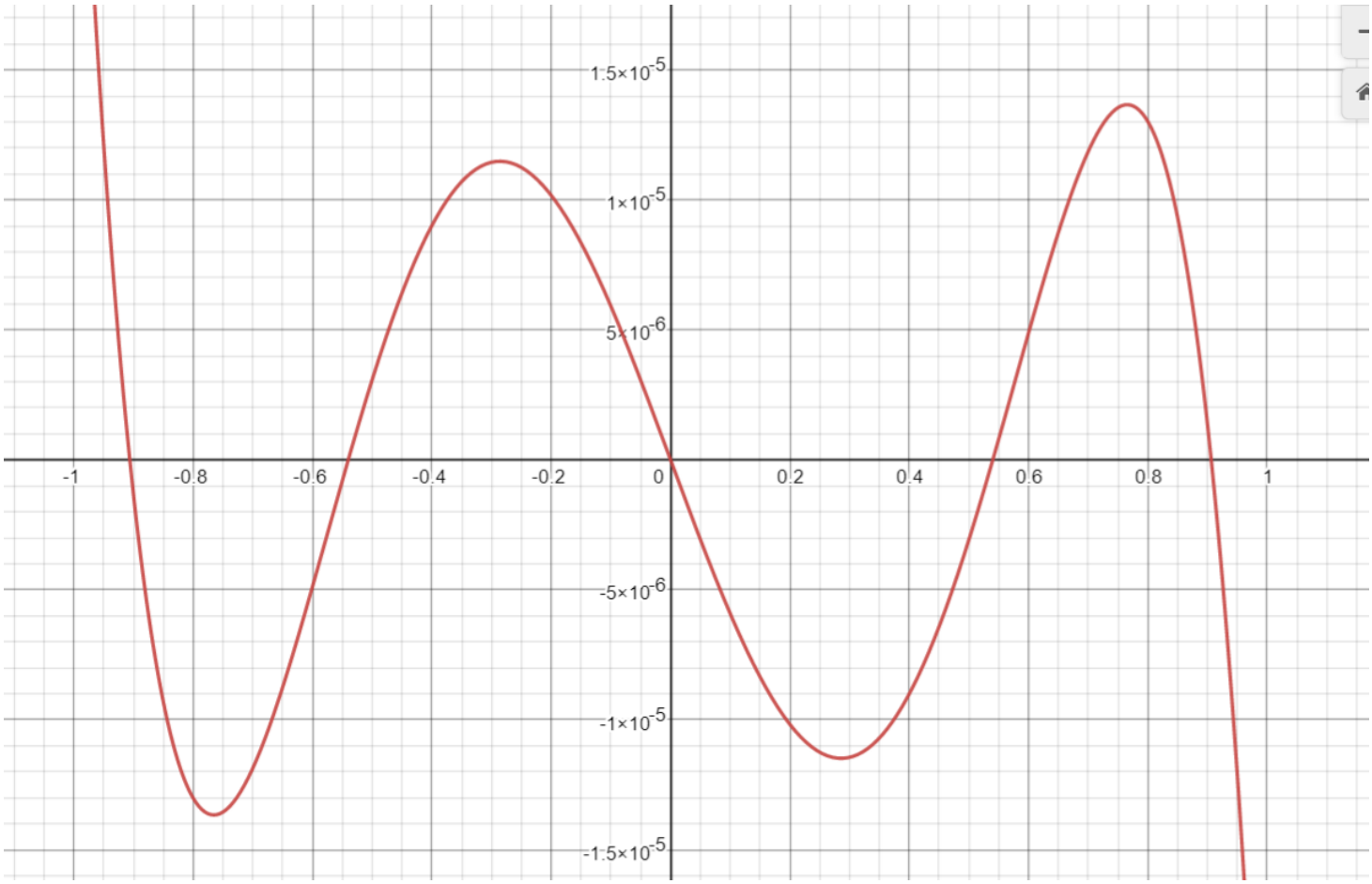
由于$a_k^*=\frac{(f,P_k)}{(P_k,P_k)}$

$$
\Rightarrow\begin{cases}a_0^*=0\\a_1^*=0.487611\\a_2^*=0\\a_3^*=-0.008218\end{cases}
$$

代入得到

$$
\begin{aligned}
S_3^*(x)&=0.487611x-0.008218(2.5x^3-1.5x)\\
&=0.499938x-0.020545x^3
\end{aligned}
$$

误差图形如图



均方误差$||\delta||^2 = \sqrt{||f||_2^2 - \sum_{k=0}^{3} a^k(f, P_k)} \approx 0.000247$

P78：第22题

$\Phi = \{1, x^2\}$，设权函数$w(x) \equiv 1$，故

$$\begin{cases} (1,1) = 5 \\ (1, x^2) = 19^2 + 25^2 + 31^2 + 38^2 + 44^2 = 5327 \\ (x^2, 1) = 5327 \\ (x^2, x^2) == 19^4 + 25^4 + 31^4 + 38^4 + 44^4 = 7277699 \end{cases}$$

$$\begin{cases} (f, 1) = \sum_i y_i = 271.4 \\ (f, x^2) = \sum_i y_i x_i^2 = 369321.5 \end{cases}$$

故求解如下方程组即可：

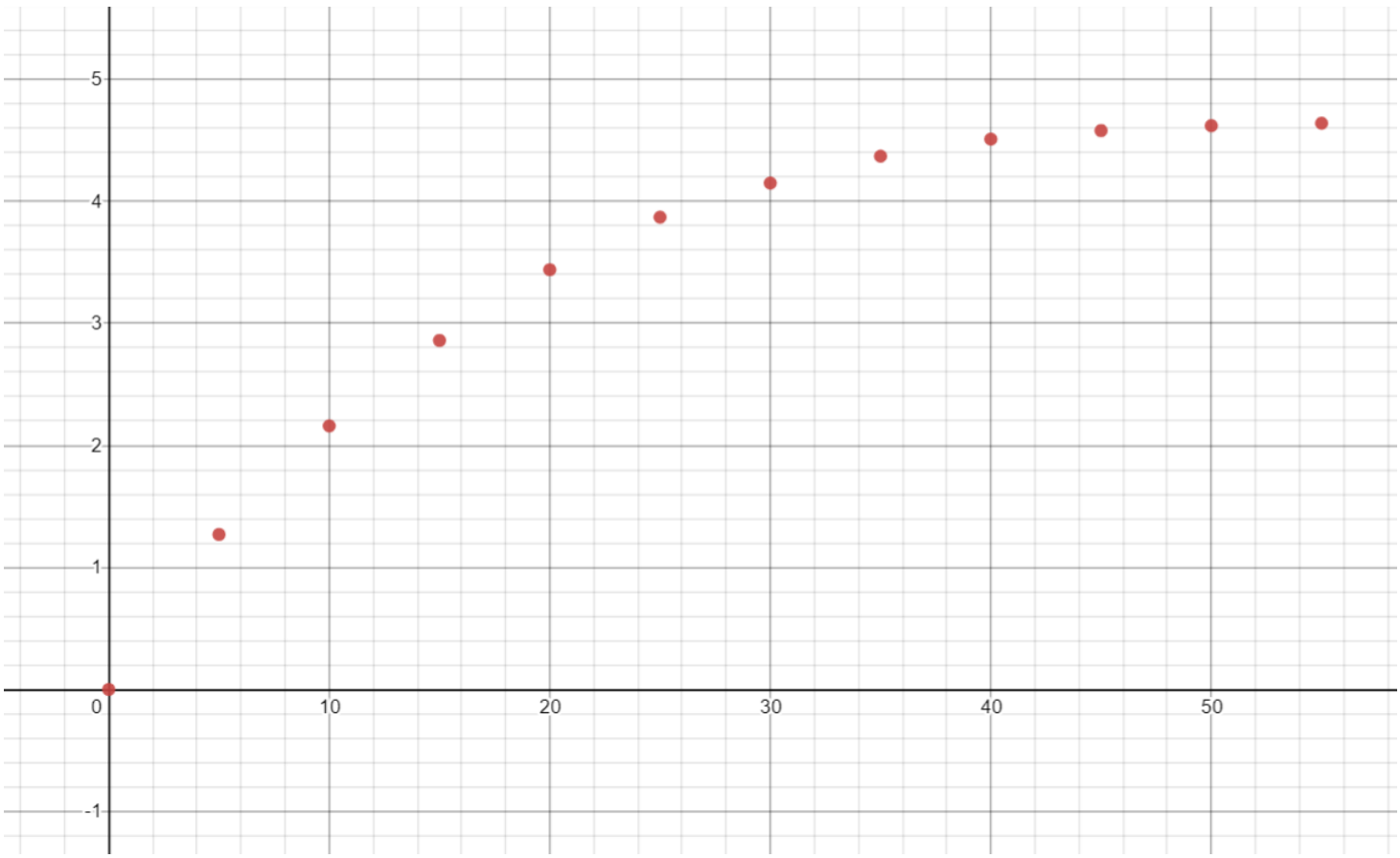
$$\begin{cases} 5a + 5327b = 271.4 \\ 5327a + 7277699b = 369321.5 \end{cases}$$

最终解得$a = 0.972605, b = 0.050035$
经验公式为$y = S(x) = 0.972605 + 0.050035x^2$
均方误差

$$\delta = \sum_i (y_i - S(x_i))^2 \approx 0.1226$$

P79：第24题

先描点画图，图片如下



观察图形，选取$y = ae^{-\frac{b}{t}}$进行拟合
$y = ae^{-\frac{b}{t}} \Rightarrow lny = lna - b \cdot \frac{1}{t}$
令$\bar{y} = lny, A = lna, x = \frac{1}{t}$，此时为利用$(x_i, \bar{y}_i)$拟合$\bar{y} = A - bx$
$\Phi = \{1, x\}$，取权函数$w(x) \equiv 1$，不考虑点$(0, 0)$
$(1, 1) = 11, (1, x) = (x, 1) = \sum_i \frac{1}{t_i} = 0.603975$
$(x, x) = \sum_i \frac{1}{t_i^2} = 0.062321$

$(f,1)=\sum_i lny_i=13.639649$
$(f,x)=\sum_i \frac{1}{t_i}lny_i=0.530330$
故求解如下方程组即可：

$$\begin{cases}11A-0.603975b=13.639649\\0.603975A-0.062321b=0.530330\end{cases}$$

解得$A=1.651562,b=7.496221$
故$lny=1.651562-7.496221\frac{1}{t}$

$$y=5.215119e^{-\frac{7.496221}{t}}(*10^{-4})$$

P104：第1题(3)

先令$f(x)=1$，有$\int_{-1}^{1}f(x)dx=[f(-1)+2f(x_1)+3f(x_2)]/3$
再分别令$f(x)=x.f(x)=x^2$，有：

$$\begin{cases}0=-1+x_1+3x_2\\2=1+2x_1^2+3x_2^2\end{cases}$$

解得

$$\begin{cases}x_1=-0.289898\\x_2=0.626599\end{cases}\text{或}\begin{cases}x_1=0.689898\\x_2=-0.126599\end{cases}$$

对于任意一组参数，都有：$\int_{-1}^{1}x^3dx\neq(-1+2x_1^3+3x_2^3)/3$，故该公式具有2次代数精度
参数即为上述两组x_1,x_2

P104：第2题(1)

步长$h=1/8$
梯形公式：$T_8=\sum\limits_{k=0}^{7}h\cdot\frac{f(kh)+f(kh+h)}{2}=0.1114024$

Simpson公式：$S_8=\sum\limits_{k=0}^{7}\frac{h}{6}\cdot[f(kh)+4f(kh+0.5h)+f(kh+h)]=0.1115718$

P105：第7题(题目改成课程PPT上内容)

复化梯形法的余项$|R_T| = |-\frac{b-a}{12}h^2 f''(\eta)| = |-\frac{1}{12}\frac{1}{n^2}f''(\eta)| \le \frac{1}{12n^2}e \le 0.5*10^{-5}$
由$\frac{1}{12n^2}e \le 0.5*10^{-5} \Rightarrow n \ge 212.849$
故复化梯形法要求$n = 213$及以上，即至少把区间分成213等分

复化Simpson法的余项$|S_T| = |-\frac{b-a}{180*2^4}h^4 f^{(4)}(\eta)| = |-\frac{1}{180*2^4}\frac{1}{n^4}f^{(4)}(\eta)| \le \frac{1}{180*2^4 n^4}e \le 0.5*10^{-5}$
由$\frac{1}{180*2^4 n^4}e \le 0.5*10^{-5} \Rightarrow n \ge 3.707$
故复化Simpson法要求$n = 4$及以上，及至少把区间分成4等分(但实际上还需要每个小区间中点的函数值，理解为8等分也可以)

P105：第11题

(1)列出T数表
$T_0^{(0)} = 1.3333333$
$T_0^{(1)} = 1.1666667, T_1^{(0)} = 1.1111111$
$T_0^{(2)} = 1.1166667, T_1^{(1)} = 1.10000000, T_2^{(0)} = 1.0992593$
$T_0^{(3)} = 1.1032107, T_1^{(2)} = 1.0987253, T_2^{(1)} = 1.0986403, T_3^{(0)} = 1.0986305$
$T_0^{(4)} = 1.0997677, T_1^{(3)} = 1.0986200, T_2^{(2)} = 1.0986130, T_3^{(1)} = 1.0986126, T_4^{(0)} = 1.0986125$

最终，$I = 1.0986125$

(2)首先把积分区间变化到$[-1,1]$，令$y = x+2$，$x \in [-1,1]$
此时$\int_1^3 \frac{1}{y}dy = \int_{-1}^1 \frac{1}{x+2}dx$

先用三点高斯：$P_3(x) = (5x^3 - 3x)/2$
其零点为$0, \frac{\sqrt{15}}{5}, -\frac{\sqrt{15}}{5}$
故$\int_{-1}^1 \frac{1}{x+2}dx \approx A_0 f(0) + A_1 f(\frac{\sqrt{15}}{5}) + A_2 f(-\frac{\sqrt{15}}{5})$
令其对$f(x) = 1, x, x^2$准确成立，求得A_0, A_1, A_2最终带入得到

$$\int_1^3 \frac{1}{y}dy = \int_{-1}^1 \frac{1}{x+2}dx \approx 1.0980393$$

再用五点高斯：$P_5(x) = (63x^5 - 70x^3 + 15x)/8$
其零点为$0, 0.9061798, -0.9061798, 0.5384693, -0.5384693$

故$\int_{-1}^{1}\frac{1}{x+2}dx \approx A_0f(x_0)+A_1f(x_1)+A_2f(x_2)+A_3f(x_4)+A_4f(x_4)$
令其分别对$f(x)=1,x,x^2,x^3,x^4$准确成立，求得A_0,A_1,A_2,A_3,A_4，最终带入得到

$$\int_1^3\frac{1}{y}dy=\int_{-1}^{1}\frac{1}{x+2}dx\approx 1.0986093$$

(3)把$[1,3]$四等分然后各个区间用Gauss公式再求和，仅给出求第一个小区间的具体过程，另外三个方法一样
$I_1=\int_1^{1.5}\frac{1}{y}dy=\int_{-1}^{1}\frac{0.5}{2.5+0.5t}dt$
$P_2(x)=0\Rightarrow x=1/\sqrt{3},-1/\sqrt{3}$
故$\int_{-1}^{1}\frac{0.5}{2.5+0.5t}dt\approx A_1f(1/\sqrt{3})+A_2f(-1/\sqrt{3})$
令上式对$f(t)=1,t$准确成立，解得$A_1=A_2=1$
故

$$I_1\approx\frac{0.5}{2.5+0.5/\sqrt{3}}+\frac{0.5}{2.5-0.5/\sqrt{3}}\approx 0.4054054$$

类似的，有$I_2\approx 0.2876712,I_3\approx 0.2231405,I_4\approx 0.1823204$

故$I=I_1+I_2+I_3+I_4\approx 1.0985375$